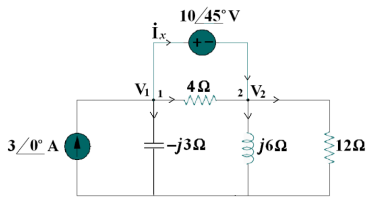


课堂笔记整理4.28

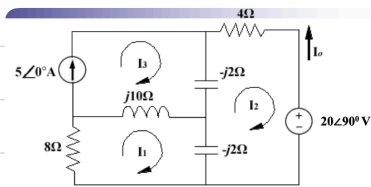
堂练1:



列KCL:
$$\begin{cases} 3\angle 0^\circ = \frac{V_1}{-j3} + \frac{V_1 - V_2}{4} + I_x \\ \frac{V_1 - V_2}{4} + I_x = \frac{V_2}{j6} + \frac{V_2}{12} \\ V_1 - V_2 = 10\angle 45^\circ \end{cases}$$

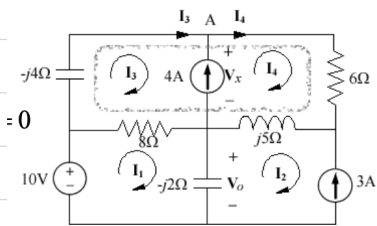
可解得 $V_1 = 25.78\angle -70.48^\circ \text{ V}$
 $V_2 = 31.41\angle -87.18^\circ \text{ V}$

堂练2:



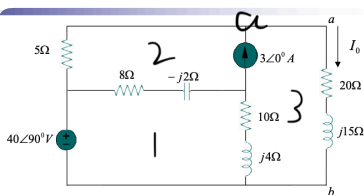
对网孔1, $(8 + j10 - j2)I_1 - (-j2)I_2 - j10I_3 = 0$
 对网孔2, $(4 - j2 - j2)I_2 - (-j2)I_1 - (-j2)I_3 + 20\angle 90^\circ = 0$
 对网孔3, $I_3 = 5$
 则有 $I_2 = 6.12\angle -35.22^\circ \text{ A}$, $I_0 = -I_2 = 6.12\angle 144.78^\circ \text{ A}$

堂练3:



对网孔1: $-10 + (8 - j2)I_1 - (-j2)I_2 - 4I_3 = 0$
 对网孔2: $I_2 = -3$
 对广义网孔: $\begin{cases} (8 - j4)I_3 - 8I_1 + (6 + j5)I_4 - j5I_2 = 0 \\ I_4 = I_3 + 4 \end{cases}$
 $I_1 = 3.618\angle 274.5^\circ \text{ A}$, $V_0 = -j2(I_1 - I_2) = 9.756\angle 222.32^\circ \text{ V}$

堂练4:



$Z_N = 5\Omega$, 网孔2,3为广义网孔,
 对网孔1, $-j40 + (18 + j2)I_1 - (8 - j2)I_2 - (10 + j4)I_3 = 0$
 对广义网孔, $(13 - j2)I_2 + (10 + j4)I_3 - (18 + j2)I_1 = 0$
 节点a: $\begin{cases} I_3 = I_2 + 3 \\ I_N = I_3 = 3 + j9 \text{ A} \end{cases}$
 则 $I_0 = \frac{5}{5 + j20 + j15} I_N = \frac{3 + j9}{j5 + j3} = 1.165\angle 38.48^\circ \text{ A}$